

EXISTENCE OF BISTABILITY AND CORRELATION WITH ARRHYTHMOGENESIS IN SHEEP ATRIA

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ABSTRACT

This experimental study investigates the response of sheep atria to periodic pacing. In 23 trials over 6 animals, a window of bistability was observed wherein for a range of pacing periods the atria responded 1:1 or 2:1 depending on whether pacing period was increasing or decreasing. In 18 trials, non-paced activations were also observed. In 17 of those 18 trials, the non-paced activations occurred at pacing periods within or no more than 30 ms away from the window of bistability. These results suggest bistability exists in cardiac tissue and is correlated with arrhythmogenesis.

INTRODUCTION

Understanding the underlying dynamics of cardiac tissue is important in its own right and is vital to the development of cardiac control techniques. Pacing is widely used to investigate cardiac dynamics. At longer pacing periods cardiac tissue responds 1:1 (one activation for every stimulus), while at shorter pacing periods it responds 2:1 (one activation for every other stimulus). As observed in frogs [1], the pacing periods at which the response switches between 1:1 and 2:1 depends on whether pacing period is increasing or decreasing. The 1:1-to-2:1 transition occurs at a shorter pacing period than the 2:1-to-1:1 transition. This hysteretic behavior yields a window of bistability wherein 1:1 and 2:1 responses coexist at the same pacing period. This phenomenon has been found in other experiments and simulations but appears to have never been investigated *in vivo* in whole mammalian hearts.

METHODS

A 528 electrode mapping/pacing plaque was affixed to the anterior free wall of the right atrium in 6 sheep. Cathodal pacing stimuli greater than two times diastolic threshold were delivered to the center of the plaque. Pacing period (T_p) was decreased from 275 ± 25 ms to 75 ± 25 ms and then increased to 230 ± 70 ms in increments of 5 or 10 ms. Monopolar extracellular electrograms were recorded using a 528 channel mapping system. They were digitized at 2 kHz, filtered, differentiated, and displayed visually on an array corresponding to the plaque. Animations of the array showing activation fronts were analyzed to characterize cardiac response, i.e., 1:1, 2:1, non-paced activations or combinations thereof.

RESULTS

Hysteresis and bistability were observed in all 6 sheep. In all cases, the 1:1-to-2:1 transition occurred at a lower T_p than the 2:1-to-1:1 transition. Figure 1 summarizes the results and clearly shows the existence of hysteresis and

bistability. At T_p 's away from the window of bistability, the tissue responds clearly 1:1 or 2:1. However, in most trials, non-paced activations similar to reentrant activations and rotors were observed at T_p 's close to and within the window of bistability. Figure 1 shows that non-paced activations were highly correlated with the window of bistability with the exception of animal 6. Specifically, non-paced activations occurred at T_p 's within the window of bistability or no more than 30 ms above or below the window of bistability. Non-paced activations which occurred at 75, 80, and 85 ms in animal 6 were most likely correlated with a switch to a 3:1 response which occurred at 70 ms.

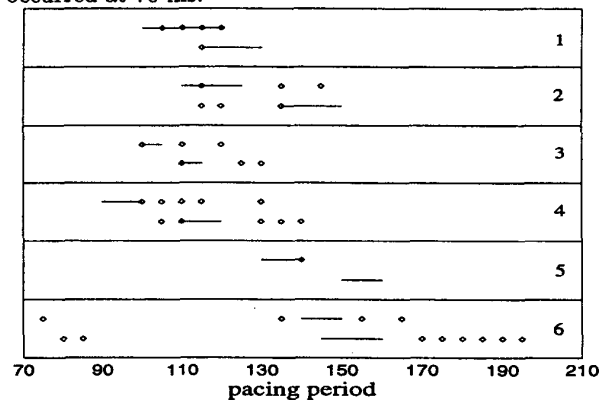


Figure 1. Transitions and non-paced activations in 6 animals. Upper line: range of 1:1-to-2:1 transitions. Lower line: range of 2:1-to-1:1 transitions. $\diamond$: non-paced activations.

DISCUSSION

A window of bistability appears to exist in sheep atria and was found to be correlated with observed non-paced activations. Therefore, bistability may be a dynamical state responsible for arrhythmogenesis and should be considered in the development of cardiac control techniques.

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REFERENCES

- [1] G. Martin Hall, Sonya Bahar, Daniel J. Gauthier, "Prevalence of rate-dependent behaviors in cardiac muscle," *Phys. Rev. Lett.*, in press.